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CO2 - ASSESSMENT TOOL - 01

CSA1610

Data Warehousing and Data Mining

Q1) @ Data Mining :-

Data Mining is the process of extracting useful patterns, relationships and knowledge from large amounts of data.

Any two basic Data Mining tasks:-

- ① classification - Assigns data into predefined categories.
- ② clustering - Groups similar data items together.

KDD steps:-

- ① Data selection
- ② Data cleaning

Role of Data processing

- * Removes errors & Missing values
- * Increases accuracy

⑥ Data cleaning :-

It is the process of removing errors and Missing values.

Two Quality Data Issues-

- * Missing values.
- * Noisy data.

Handling Missing values-

- * Replace with mean or Median
- * Delete incomplete records.



Handling Noisy data:-

- * Binning
- * Regression

Q2. (a) Data Integration:-

It combines data from different sources into one data set.

Challenges:-

- * Data redundancy.
- * Data Inconsistency.

Handling:-

- * Remove duplicate records.
- * Standardize data formats.

(b) Covariance Analysis

Covariance measures how two variables change together.

Two Covariance Formulas-

Population Covariance:-

$$\text{Cov}(X, Y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$

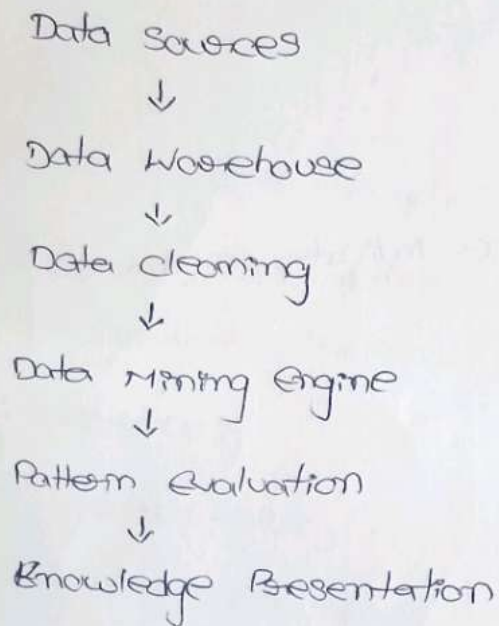
Sample Covariance:-

$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

a) From given Condition -

The Covariance is positive, therefore, stock A & stock B rise together and are not independent.

b) Architecture of Data Mining system



Components :-

- * Data Sources - store data
- * Data Warehouse - collects data
- * Data cleaning - Remove errors
- * Data Mining Engine - Finds pattern
- * Pattern Evaluation - Select useful patterns
- * Knowledge Presentation - Display results.

④ a) Given:

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = 6.2$$

$$\text{Mean}(Y) = 5.8$$

$$\text{Covariance} = -0.45$$

Interpretation:- Negative Covariance indicates an inverse relationship.

b) Major steps-

- ① Data cleaning
- ② Data Integration
- ③ Data Transformation
- ④ Data Reduction
- ⑤ Data Discretization

Pipeline:-

Missing values $\rightarrow$ Remove duplicates $\rightarrow$ Handle outliers $\rightarrow$

Format data $\rightarrow$ Normalize $\rightarrow$ Reduce data.

Q) Equal Frequency Binning (3 Bins)

Bin 1:-

13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2:-

22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3:-

33, 35, 35, 36, 40, 45, 46, 52, 70

B) Data Discretization-

It converts continuous data into intervals (bins)

Importance:-

- * Reduces noise
- * Improves accuracy
- * Simplifies analysis

Equal-Width Binning:-

- * Equal interval size

Equal-Frequency binning:-

- * Equal number of values in each bin.

Comparison:-

① Equal-Width :- Same interval size.

② Equal-Frequency :- Same no. of records.